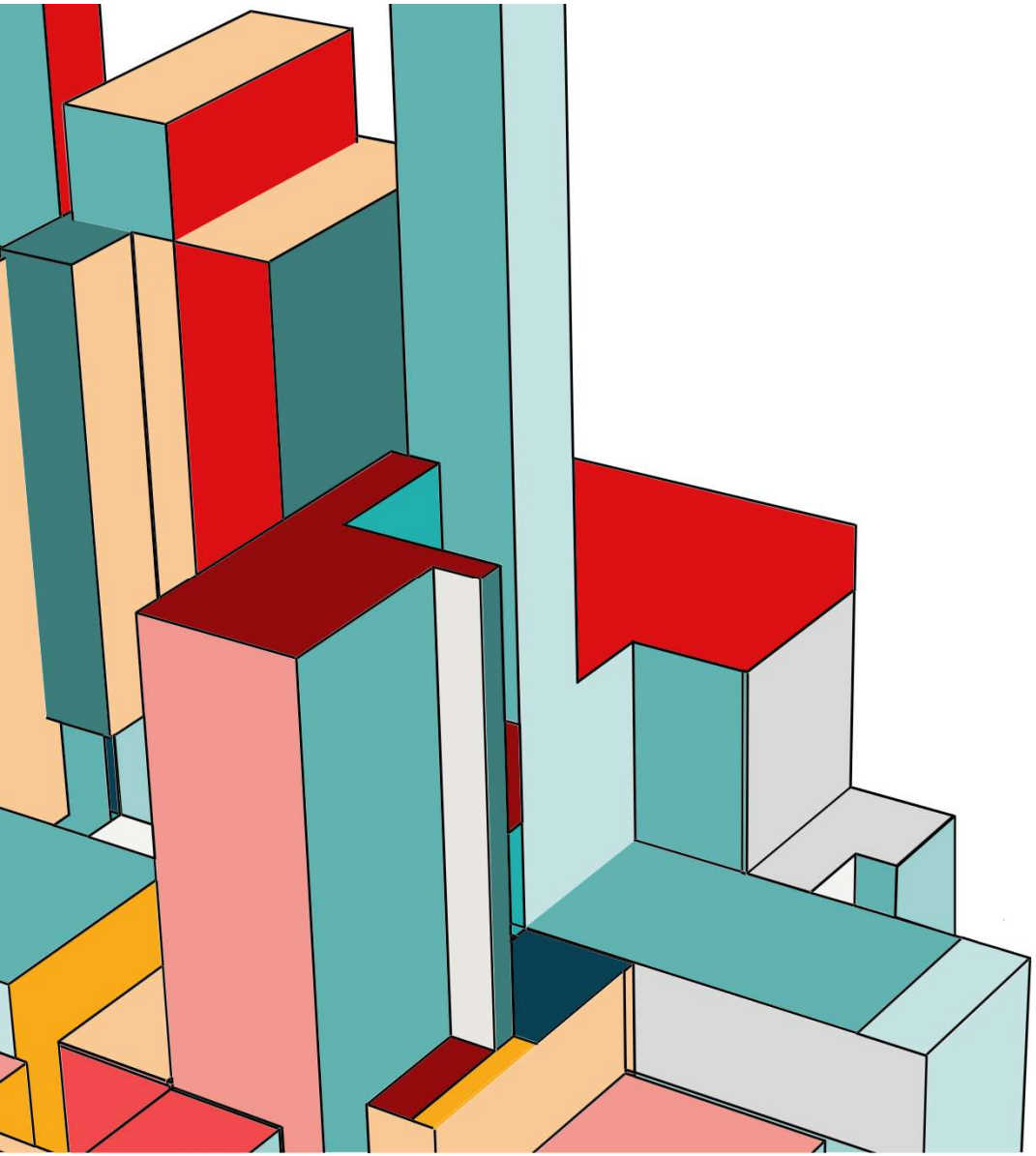


CHECK IN-2

Ahad Nardekar

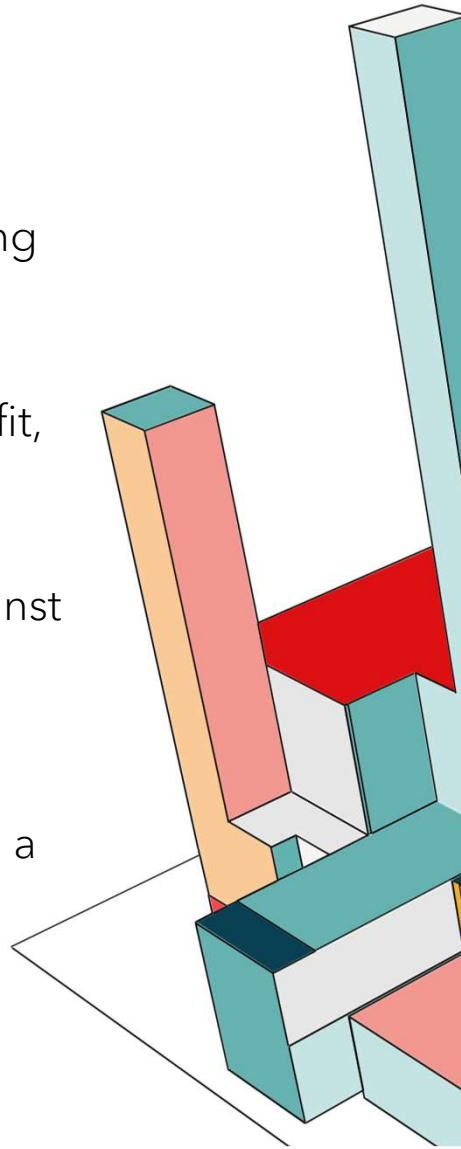
NUS National University of Singapore

NUS Cognition X Lab

Data-informed Smart Built Environment for Sustainability

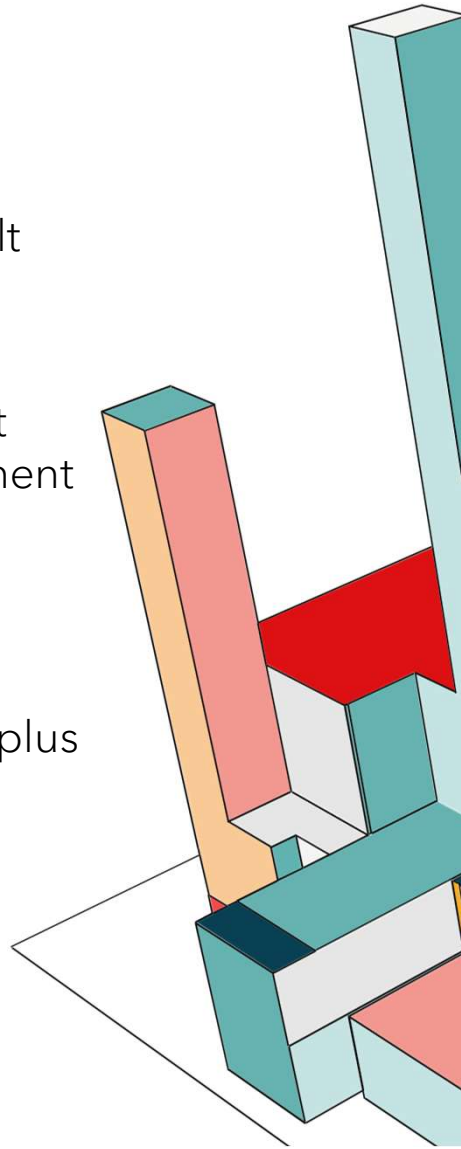
WORK SO FAR

- Built and maintained a structured literature master sheet with full screening and detailed-review columns
- Produced a model-family decision sheet comparing six families (lexicon, classical ML, fine-tuned encoders, generative LLMs, RAG, hybrid) on task fit, data needs, strengths, and weaknesses, anchored on one running report example.
- Mapped the four reporting regimes (GRI, SASB, TCFD, ISSB/IFRS S2) against Scope 1, 2, and 3 requirements to separate mandatory from self-selected disclosure. [partly Raquel]
- Built a working understanding of evaluation metrics (precision, recall, F1, accuracy) with a worked greenwashing-detector example, ready for when a classifier is validated.



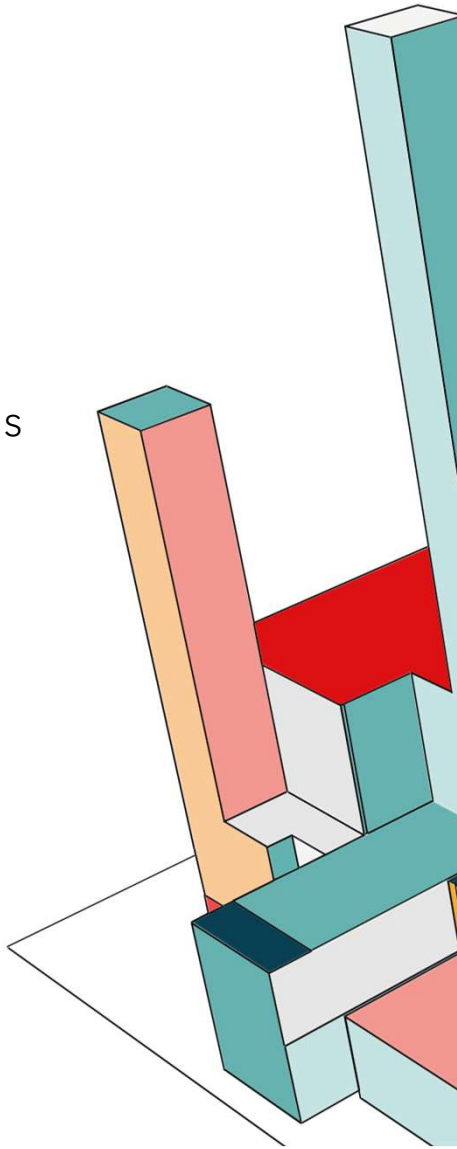
CONCLUSIONS FROM THE WORK SO FAR

- The decoupling result needs both halves, and the talk half is far more built than the walk half, so emissions and green capex extraction are now the bottleneck.
- The hard part of the field is specificity, not classification: category tasks hit near-perfect F1 with simple baselines, while specificity sits at 0.17 agreement (P2), so the project's real risk is in the scoring step.
- Classification, extraction, scoring, similarity, and sequence labelling are genuinely different jobs and each deserves its own model choice.
- For a two-month replication package, a simple transparent stack (lexicon plus classical ML plus light LLM extraction) is cheap, reproducible



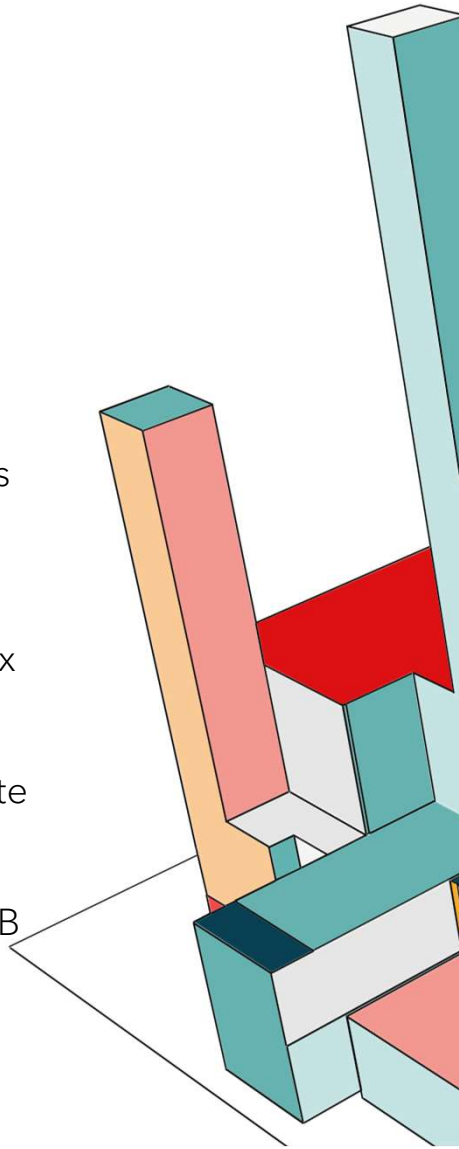
CONCLUSIONS FROM THE WORK SO FAR

- LLM zero-shot extraction (the P10 route) is the most flexible option when labels are scarce, at the cost of reproducibility and hallucination risk that has to be validated.
- Cosine similarity to TCFD categories captures which category a segment is in, not how committed it is, so it cannot be the TalkScore engine alone. Need to add confidence score as well.
- RAGs can be used for extracting relevant texts from the long pdfs
- No verified greenwashing dataset exists anywhere in the field (P2), so validation leans on proxy tasks and third-party signals and the project cannot claim confirmed greenwashing detection.

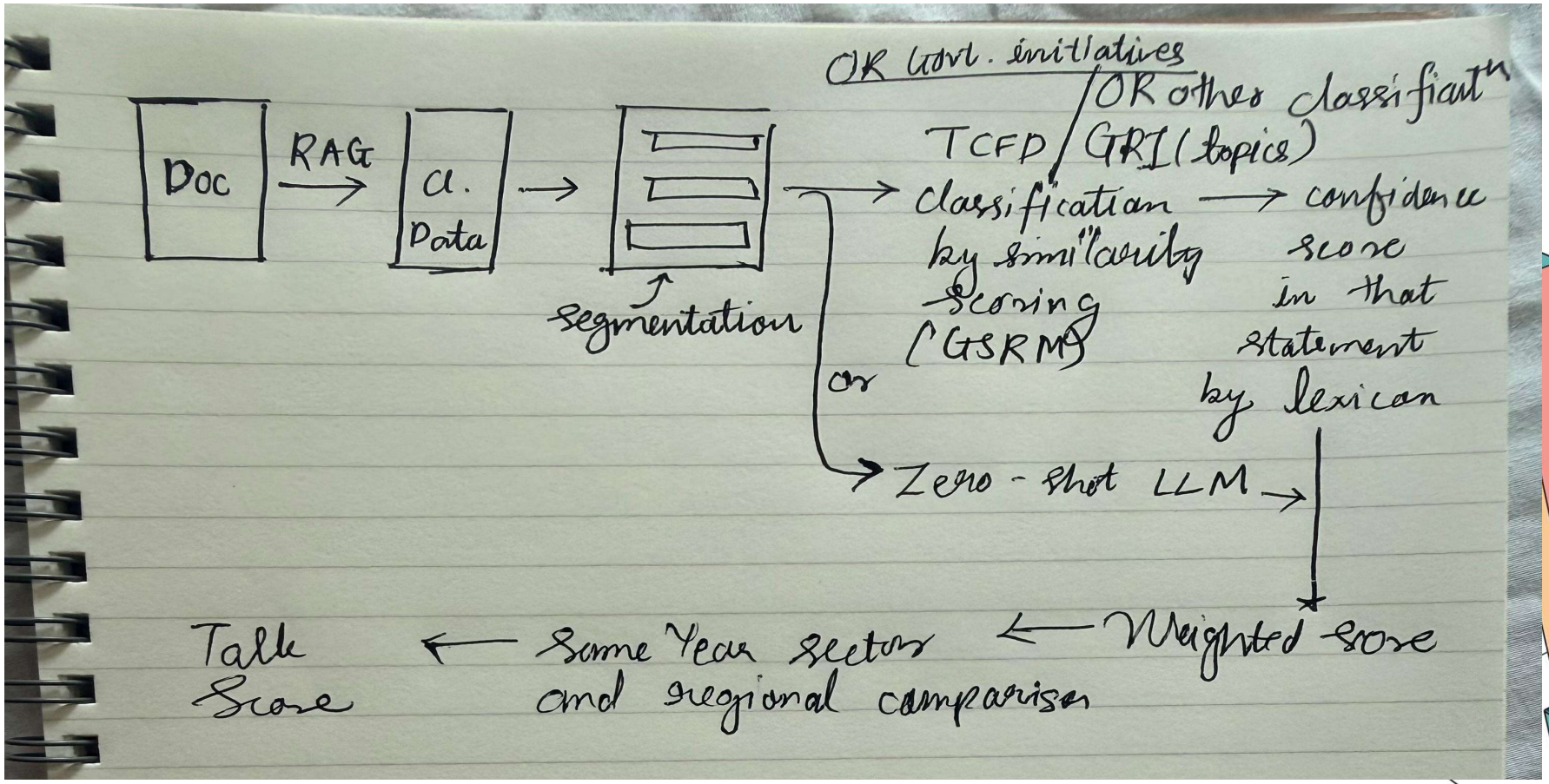


LIMITATIONS AND DEAD ENDS

- TalkScore's exact output is still undefined (single score, per-theme vector, specificity rating, or compliance-gap flag against the ISSB floor), and model choice can't be finalised until it is.
- No verified ground truth exists, so there is no clean label to train or test on, so use the manual gap procedure as silver-standard labels on a small sample and state plainly it is a proxy.
- The emissions and financial side is barely built and no firm-level dataset is defined, so lock a small fixed sample of HK/SG firms first and hand-build their emissions and capex series.
- Cross-year panels are messy because firms only moved to ISSB recently and may restate baselines, so start single-year and treat time as a later extension, from Raquel's work.
- ClimateBERT may misfit ISSB content because it is TCFD-trained, so test it on a few ISSB reports before committing and keep an LLM route as backup, from Raquel's work.



PROPOSED MODEL



SOME QUESTIONS

- What exactly should TalkScore output: one commitment score per firm, a score per ISSB theme, a specificity rating per pledge, or a flag for falling below the ISSB floor?
- For the action side, is realised emissions trajectory enough, or do you want green capex and operational commitment included given how hard green capex is to extract?
- With no verified Is single-year ISSB analysis acceptable as the main result, with cross-year as an extension, given firms only adopted ISSB recently?
- Should peer comparison (firm versus sector norm) be part of the output, given P10's similarity machinery makes it cheap?
- For green capex that cannot be isolated, is marking firms unknown and excluding them acceptable, or do you want a narrative proxy?

